

Department of Computer Engineering

National Institute of Technology Kurukshetra

B.Tech (IT and MC), ITPC-206/DBMS, Mid Term (I) Exam, 4rth Sem Sem, AY. 2025-26

Time Allowed: 50 Mins

Max. Marks: 15.00

Date of Exam: 20 Feb, 2026

Roll No: _____

Note: All the questions and their parts are compulsory; there is no choice. Answer each part of a question in the same place.

Q1. Answer the following briefly:

- (a). "City Hospital" records every patient's temperature, blood pressure, and medication times as raw data in its database. The hospital's analytics system processes this data to create an information dashboard for the head nurse. This dashboard highlights patients with abnormal vital signs and predicts which patients are at risk of readmission. *Identify two examples of data and two examples of information from the above.*
- (b). Is it possible to successfully map a binary $M:N$ relationship type without requiring a new relation? Why or why not?
- (c). How is data retrieval different from information retrieval?

Q2. (a) As the new IT coordinator for a small university, you observe that the Registrar's office maintains student data in an Excel spreadsheet, the Sports Department keeps a separate CSV file for team registrations, and the Library manages its own list of students with borrowing privileges in an Access database. A student, Alice Johnson, gets married and changes her last name to "Smith." Although she submits the change to the Registrar's office, who updates their spreadsheet, the Sports Department and Library are not notified of this change. Months later, Alice is unable to check out a book because the library system still has her under "Johnson," and her sports eligibility is questioned due to the name mismatch. *Explain how the lack of a centralized database system led to this situation. What specific problems of file-based systems does this scenario illustrate, and how would a database system have prevented this inconsistency?*

- (b) Composite and multivalued attributes can be nested to any number of levels. Suppose we want to design an attribute for a STUDENT entity to keep track of previous college education. Such an attribute will have one entry for each college previously attended, and each such entry will be composed of college name, start and end dates, degree entries (degrees awarded at that college, if any), and transcript entries (courses completed at that college, if any). Each degree entry contains the degree name and the month and year the degree was awarded, and each transcript entry contains a course name, semester, year, and grade. *Design an attribute to hold this information.*

Q3. (a) The National Election Management System is designed to handle the complex processes of electoral administration in a structured manner. Citizens are registered as voters and uniquely identified by a voter ID. Each voter belongs to a specific constituency, and constituencies are organized under states. Candidates contest elections within constituencies, with the condition that every candidate must also be a registered voter. Votes are recorded electronically, and each vote is associated with the voter, the candidate, and the constituency. Political parties play a crucial role by nominating candidates, while also allowing for the possibility of candidates switching parties over time. Campaign events are organized, where multiple candidates interact with and address voters. A recursive relationship exists where candidates may endorse or support other candidates. Furthermore, voters can be classified under specialization as Resident voters or Non-Resident Indians (NRIs), reflecting different participation categories in the electoral process. *Design an Entity-Relationship Diagram (ERD), with appropriate entities (strong & weak), attributes (PK, derived, multi, & complex), relationships (recursive, etc), cardinalities (Min & Max) and participation construct (Total, Partial).*

- (b) *Design an EER schema for a database application that you are interested in. Specify all constraints that should hold on the database. Make sure that the schema has at least five entity types, four relationship type, a weak entity type, and two superclass/subclass relationships (for each Overlap and Disjoint type).*